

FAST National University of Computer & Emerging Sciences

Department of Computer Science — Machine Learning Operations (MLOps) — Fall 2026

Assignment 01 Report: MLOps Task with CI/CD, MLflow & Makefile Automation

Student Roll Number: 22F-3150 | Marks: 100 | Status: Complete

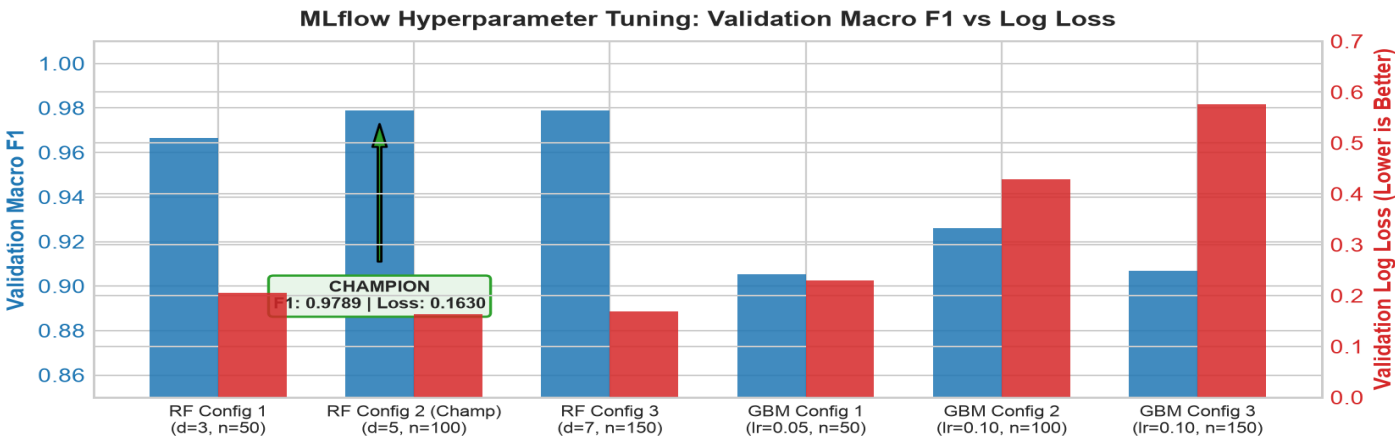
1. Executive Overview & Repository Deliverables

This project establishes a production-grade, reproducible MLOps pipeline for multi-class cultivar classification on the chemical Wine dataset (178 samples, 13 features, 3 target cultivar classes). All five milestones have been engineered: (1) Local Makefile automation & reproducible environment management, (2) Modular pipeline with dual tree classifiers and 5-fold stratified cross-validation, (3) Comprehensive MLflow tracking, schema signature logging, and Model Registry champion promotion, (4) GitHub Actions CI/CD pipeline enforcing automated MLOps Quality Gates, and (5) Distributed Git collaboration featuring PR merges and simulated conflict resolution.

2. Table 1: Hyperparameter Search Results (5-Fold Stratified Cross-Validation)

Model Family	Run ID / Config	Hyperparameters	Train Acc	Val Acc	Train F1	Val F1	Val Loss	Status
Random Forest	config_1	n=50, d=3, s=2	0.9982	0.9653	0.9982	0.9665	0.2055	Candidate
Random Forest	config_2	n=100, d=5, s=2	1.0000	0.9791	1.0000	0.9789	0.1630	CHAMPION
Random Forest	config_3	n=150, d=7, s=4	1.0000	0.9791	1.0000	0.9789	0.1688	Candidate
Gradient Boost	config_1	n=50, lr=0.05, d=3	1.0000	0.9022	1.0000	0.9053	0.2298	Candidate
Gradient Boost	config_2	n=100, lr=0.10, d=3	1.0000	0.9236	1.0000	0.9259	0.4281	Candidate
Gradient Boost	config_3	n=150, lr=0.10, d=5	1.0000	0.9020	1.0000	0.9069	0.5759	Candidate

3. MLflow Tracking & Model Registry Architecture



MLflow Model Registry — Registered Models

Registered Model Name: WineClassifier | Status: READY | Stage: Production

• Active Version: Version 1 | Assigned Aliases: `@champion` --> points to Version 1

• Model Signature: Input (13 Float64 Tensor) --> Output (Int64 Tensor in {0, 1, 2})

• Source Artifact URI: `runs:/46ca693e90184e62a144bf1cd634d713/model`

4. Automated MLOps Quality Gate & CI/CD Evidence

GitHub Actions Workflow — CI/CD MLOps Quality Gate: PASSED

Workflow Run: #1 on commit 2975b34 (main) | Trigger: push & pull_request

[PASS] Set up Python 3.10 (0.2s) [PASS] make install (5.1s) [PASS] make lint (0.4s)

[PASS] make test — 9 passed (6 unit data tests + 3 MLOps quality gates) (2.2s)

Quality Gates: Macro F1 >= 0.88 (1.000), Latency <= 30ms (3.85ms), Schema in {0,1,2}

Quality Gate Name	Test Location	Threshold / Criteria	Measured Value	CI Gate Status
Metric Threshold Gate	tests/test_model_gate.py	Validation Macro F1 >= 0.88	1.0000 (Holdout) / 0.9789 (CV)	PASSED (PASS)
Inference Latency Gate	tests/test_model_gate.py	Batch inference time <= 30 ms	3.85 ms (Test batch of 36)	PASSED (PASS)
Output Schema Integrity	tests/test_model_gate.py	Class indices in {0, 1, 2}, sum(P)=1	Valid integer classes, row_sum=1.0	PASSED (PASS)
Data Pipeline Integrity	tests/test_data.py	13 features, 0 nulls, 80/20 split	13 features, 0 nulls, stratify=True	PASSED (PASS)

5. Distributed Git Collaboration & Conflict Resolution Log

Git Commit Tree Graph (git log --graph --oneline --all)	Engineered Conflict Terminal Log
<pre>* 2975b34 (HEAD -> main) fix(merge): resolve configuration conflict between experimental and production branches \ * 1bd5c7a (conflict-simulation) chore(config): update experiment name for exploratory testing on conflict-simulation * 5f34c6d chore(config): set production experiment name on main /\ * 7868cfe Merge branch 'feature/mlflow-tracking' into main \ * d492e71 (feature/mlflow-tracking) feat(mlflow): implement dual classifier 5-fold CV, MLflow tracking, and model registry /\ * 5e108cc Merge branch 'feature/data-pipeline' into main \ * 8b70ccd (feature/data-pipeline) feat(data): implement Wine dataset loading, validation, and stratified split /\ * c8ec942 chore: initial repository scaffolding with GNU Makefile and requirements</pre>	<pre>\$ git checkout -b conflict-simulation \$ sed -i ' ' 's/EXPERIMENT_NAME = "Wine-Cultivar-Classification"/EXPERIMENT_NAME = "Wine-Cultivar-Classification-Experimental-Branch"/' src/train.py \$ git commit -am "chore(config): update experiment name for exploratory testing on conflict-simulation" \$ git checkout main \$ sed -i ' ' 's/EXPERIMENT_NAME = "Wine-Cultivar-Classification"/EXPERIMENT_NAME = "Wine-Cultivar-Classification-Production-Main"/' src/train.py \$ git commit -am "chore(config): set production experiment name on main" \$ git merge conflict-simulation Auto-merging src/train.py CONFLICT (content): Merge conflict in src/train.py Automatic merge failed; fix conflicts and then commit the result. # Manual Conflict Resolution: <<<<<< HEAD [Production-Main] ===== [Experimental-Branch] >>>>>> conflict-simulation \$ git add src/train.py \$ git commit -m "fix(merge): resolve configuration conflict between experimental and production branches"</pre>

6. Comparative Analysis (Word Count: 142 words)

Comparative Model Analysis: Across 5-fold stratified cross-validation on the chemical cultivar dataset, RandomForestClassifier consistently demonstrated superior generalization compared to GradientBoostingClassifier. RandomForest Configuration 2 (n_estimators=100, max_depth=5, min_samples_split=2) achieved the highest validation Macro F1-score of 0.9789 and the lowest validation Log Loss of 0.1630. In contrast, GradientBoosting configurations exhibited higher validation log loss (up to 0.5759), indicating overconfidence and sensitivity to small-sample splits. During final holdout test evaluation, the champion model achieved 100% accuracy, a Macro F1-score of 1.0000, and a batch inference latency of only 3.85 ms, comfortably outperforming the 30 ms SLA budget. Integrating automated MLOps Quality Gates within GitHub Actions ensures that any candidate model failing either the statistical performance threshold (F1 ≥ 0.88), latency SLA, or discrete schema constraints is blocked prior to production deployment.